[Poster] QR Code Alteration for Augmented Reality Interactions

Han Park
Interactive Media Laboratory
Hongik University, Seoul, Korea

Taegyu Kim
Interactive Media Laboratory
Hongik University, Seoul, Korea

Jun Park*
Interactive Media Laboratory
Hongik University, Seoul, Korea

ABSTRACT

QR code, for its recognition robustness and data capacity, has been often used for Augmented Reality applications as well as for other commercial applications. However, it is difficult to enable tangible interactions through which users may change 3D models or animations. It is because QR codes are automatically generated by the rules, and are not easily modifiable. Our goal was to enable QR code based Augmented Reality interactions. By analysis and through experiments, we discovered that some parts of a QR code can be altered to change the text string that the QR code represents. In this paper, we introduced a prototype for QR code based Augmented Reality interactions, which allows for Rubik's cube style rolling interactions.

Keywords: QR code, Interaction, Augmented Reality.

Index Terms: H.5.1 [Information Interfaces and Presentation] : Multimedia Information Systems — artificial, augmented, and virtual realities

1 INTRODUCTION

QR codes, for its capacity to contain long string patterns, are widely used in many applications. QR codes also have been used for several Augmented Reality (AR) applications and services, providing online multimedia contents [1][2].

However, a QR code itself is a static pattern; hence it is difficult to allow user interactions. If allowed, manipulation directly applied on physical markers may be an effective mean of interaction. For example, placement of a black magnet on certain areas of a marker caused conversion of character animations or property items for an AR based storyboard application [3].

One potential approach is to alter a part of a QR code to change the text string that the code represents. Unfortunately, because the QR code patterns are automatically generated along with error correction codes, it is difficult to alter QR code patterns. Random modification results in recognition failure. Therefore, for a new text string, a new QR code is generated and printed instead of reusing a previously used similar QR code.

However, in many applications, it would be beneficial if the text string can be altered accordingly based on small changes in a QR code. We may prefer situations where users are able to switch multimedia contents with one printed QR code.

Through analysis and investigation, we could reveal a specific approach of QR alteration: middle layer replacement. Confirming through experiments, we developed a prototype, which is based on Rubik's cube puzzle. Through the interaction of rolling the middle layers of a Rubik's cube, AR contents that the QR code represent could be changed.

2 QR CODE PATTERNS

QR Code is a type of matrix barcode that has a complex structure representing message data [4]. A QR code consists of finder patterns, format information, version information, timing belt, alignment pattern, and encoded information as shown in Figure 1. Area that can be used to change the text string is encoded information(data code and error check code) area. Other areas contain information for QR code detection, and should be kept unchanged.

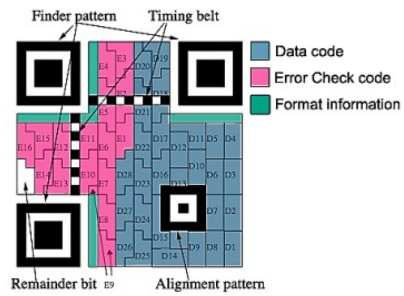

Figure 1: QR code patterns

3 QR CODE PATTERN ALTERATION

In this section, we identified the QR code areas that should be kept unchanged for successful recognition. Furthermore, we introduced a method for altering QR codes to change the representing text strings.

Because QR codes are composed of black and white bit patterns, decoded output (text string that the QR code represents) of a QR code may be changed by changing the bit patterns.

Among the QR code compartments, finder patterns, timing belt, and alignment patterns are used to detect the position of the code. Additionally, the version area contains version information. If one of these areas are altered or damaged, QR code recognition may fail.

Data patterns that include encoding information are placed in the areas starting at D2 in Figure 1. If we want to change the front part of the text string, we need to change data at D2 and D3.

However, alteration of data code only does not effect on recognized text string changes because error check code has still not been changed. Data code changes without changing the error check code was often detected as an error or even detected as the original data code.

For this reason, error check codes need to be also altered with data codes. Therefore, our altering region was determined as areas which include a part of the data code and a part of the error check code. We discovered that it is enough to alter the middle layer for partial text string changes.

Figure 2 shows an example for altering areas of a QR code. QR

* e-mail : jpark@hongik.ac.kr

IEEE International Symposium on Mixed and Augmented Reality 2014
Science and Technology Proceedings
10 - 12 September 2014, Munich, Germany
978-1-4799-6184-9/13/$31.00 ©2014 IEEE

code in Figure 2 (a) represents ‘abcdefghijklmnopqrstuvwxyz’ and QR code in Figure 2 (b) represents ‘ab44efghijklmnopqrstuvwxyz’. The changed string from ‘cd’ of (a) to ’44’ of (b) was reflected on changes at areas D3 and D4 (these areas are show in Figure 1).

QR code in Figure 2 (c) is the result of middle layer replacement. It is composed of the background region (regions outside the red rectangle region) of the QR code in Figure 2 (a) and the blue rectangle area of the QR code in Figure 2 (b). QR code (c) is based on QR code (a), but its text string is recognized as ‘ab44efghijklmnopqrstuvwxyz’ (same as QR code (b)). In this example, altering middle layer of a QR code in the horizontal direction changed the 3rd and the 4th characters of the text strings. In a similar way, vertical direction of QR code can be used to change the last two characters (at areas D25 and D26).

(a) (b) (c)

Figure 2: Middle band replacement for QR code alteration. The red region of (a) is replaced with the blue region of (b), resulting in (c).

In the previous examples, only two characters were changed. We performed several experiments of changing different numbers of characters. As long as the changed characters belong to a same middle layer (horizontal or vertical), the number of changed characters did not matter: recognition was successful regardless of the number of changed characters.

However, alteration cannot be applied in both directions simultaneously. For example, when the vertical middle layer was altered after the horizontal middle layer had been altered, QR code recognition failed.

4 QUBE AR (CUBE STYLE QR CODE AR INTERACTION)

We applied this approach of QR code alteration to a prototype which is based on Rubik’s cube puzzle. In this prototype, users were allowed to roll the middle layers to form different QR codes. We called this prototype “QubeAR” because it is a QR code in Cube shape for AR services.

The alteration of QubeAR was performed in the following way. Each face of the cube is divided into nine square regions while QR code can be also divided into nine square regions. Then, six QR codes, each divided into nine pieces, were attached on to six faces of a Rubik’s cube as shown in Figure 3 (a).

(a) (b) (c)

Figure 3: (a) QubeAR prototype (b) video of a dinosaur hatching out of its egg (c) 3D skinned model of a dinosaur

Because the middle layer of the QR code can be replaced with that of other QR codes, our intended interaction was to roll the middle layer of the Rubik’s cube. Changing middle layer of Rubik’s cube in vertical or horizontal directions, different QR code was composed.

QubeAR was applied for showing the life time and 3D models of a dinosaur. As QubeAR provides interactions in two-dimensions (horizontal and vertical), we assigned horizontal dimension to time dimension and vertical dimension to 3D model dimension. Turning the middle layer of QubeAR in the horizontal direction, a dinosaur’s life cycle was shown in images and videos.

In the beginning, dinosaurs in their eggs were shown. As the middle layer is turned, a dinosaur hatches out of its egg (Figure 3(b)). As it is turned again, a grownup dinosaur hunts another dinosaur.

Turning the middle layer in the vertical direction showed dinosaur’s 3D models in skin depth and skeleton depth (Figure 3(c)). Users were allowed to view the 3D models in different views.

QR code detection and tracking were implemented based on ZXing SDK and Qualcomm SDK [5][6].

5 CONCLUSION

QR code has several advantages: robust recognition, high data capacity, and quick response. However, in order to enrich AR and other commercial services, QR code requires interaction capabilities. Most of previous QR code researches were related to applications of utilizing QR code’s data capacity.

In this paper, we presented a prototype for QR code based tangible interactions. QubeAR is a modification of Rubik’s cube puzzle, where QR codes were attached on the faces of the cube. Based on the experiments, we discovered that the middle layers of the cube may be replaced resulting in different text strings.

Through tangible interfaces provided by this prototype, users could perform interactions of switching AR contents. According to a preliminary usability test, QubeAR scored higher in understandability and satisfaction than the traditional QR code approach.

Compared with marker-based interactions, our method is robust under partial occlusions, and may recognize a larger number of codes for employing QR codes. We expect that this prototype may be utilized for exhibitions, marketing, and education.

ACKNOWLEDGEMENTS

This work was supported by the Industrial Strategic Technology Development Program (10041656, Open Platform Development for See-Through Smart Glasses with Smart Vision) funded by the Ministry of Knowledge Economy (MKE, Korea)

REFERENCES

[1] Ruan Kong, and Hong Jeong. “An Augmented Reality System Using Qr Code as Marker in Android Smartphone”, Engineering and Technology (S-CET), 2012 Spring Congress on. IEEE, 2012.

[2] Kan Tai-Wei, Chin-Hung Teng, and Wen-Shou Chou. “Applying QR code in augmented reality applications”, Proceedings of the 8th International Conference on Virtual Reality Continuum and its Applications in Industry, ACM, 2009.

[3] Mideum Shin, Byungsoo Kim, Jun Park, “AR Storyboard: An Augmented Reality based Interactive Storyboard Authoring Tool”, IEEE and ACM International Symposium on Mixed and Augmented Reality, Oct. 6-8, 2005, Vienna, Austria, pp.198-199

[4] ISO/IEC 18004:2000 Information Technology: Automatic Identification and Data Capture Techniques (Barcode Symbology) QR Code (MOD), June 2000.

[5] http://zxing.org/w/decode.jspx

[6] https://developer.qualcomm.com/mobile-development/mobile-technologies/augmented-reality